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Palaeomagnetism of late Cretaceous sediments from southern Tibet: Evidence for the consistent palaeolatitudes of the southern margin of Eurasia prior to the collision with India

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ABSTRACT

Twenty sites were drilled in the late Cretaceous Shexing Formation for palaeomagnetic studies in the Lhasa terrane near the locality of Maxiang (29.9°N/90.7°E). The stepwise thermal demagnetizations successfully isolated high unblocking temperature characteristic directions. The tilt-corrected mean direction is D/I = 350.8°/32.1° with α_{95} = 8.1° and N = 20 sites, corresponding to a paleopole at 75.0°N, 306.7°E with A_{95} = 6.8°. Positive fold tests indicate a primary origin for the characteristic remanence. Based on previous Cretaceous data mainly from the Takena Formation and Paleocene data from the Linzizong volcanic rocks near the city of Lhasa, the latitude of the southern margin of Asia is located at about 15°N, and yields a stable position of the Lhasa terrane during Cretaceous and Paleocene. Compared with expected paleomagnetic directions from the stable India and Eurasia blocks, the collision palaeolatitude further implies the total latitudinal convergence was accommodated by 1700 ± 800 km (16.2 ± 7.6°) between southern Tibet and stable Eurasia and 1500 ± 830 km (14.4 ± 7.9°) between southern Tibet and stable India since the collision of India and Eurasia. A collision age between c. 54 and 47 Ma was determined using the results for the southern margin of Eurasia according to our new data and the extent of 'Greater India'.

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1. Introduction

The collision between the India and Eurasian continents is considered to be one of the major tectonic events of Cenozoic time. The precise timing of the initial India–Asia contact and the amount of crustal shortening remains a matter of dispute until recently. Based on different methods, the proposed initial time of collision, ranges from 34 Ma to 70 Ma (Searle et al., 1987; Rage et al., 1995; Rowley, 1998; Yin and Harrison, 2000; Aitchison and Davis, 2001; Ding, 2003; Aitchison et al., 2007, 2011; Mo et al., 2007; Xia et al., 2011). Paleomagetic data are among the robust qualitative constraints that can be used to reconstruct the block boundaries and their history. Previous paleomagnetic studies on the Cretaceous–Cenozoic units in the Lhasa block have played a key role in reconstructions of the southern margin of Eurasia block (Achache et al., 1984; Lin and Watts, 1988; Otofuji et al., 1991; Chen et al., 1993; Sun et al., 2008, 2010; Chen et al., 2010; Dupont-Nivet et al., 2010a; Liebke et al., 2010; Lippert et al., 2010; Tan et al., 2010). However, large uncertainties remain in the existing palaeomagnetic data, which have been used to constrain the palaeolatitude of the southern margin of Asia prior to the collision. The Cretaceous sedimentary rocks yield a wide range of palaeolatitude estimates from 7°N to 32°N (Westphal and Pozzi, 1983; Achache et al., 1984; Lin and Watts, 1988; Chen et al., 1993; Tan et al., 2010). This large disparity may be partly attributed to low-latitude bias due to palaeomagnetic inclination shallowing during deposition and compaction of sediments (Tauxe, 2005; Tan et al., 2010). Several paleomagnetic studies have been carried out from the Tertiary volcanic rocks from the Lhasa terrane to avoid inclination shallowing. However, these data still are disputed, primarily because part of these studies are based on too few flows to average secular variation (Achache et al., 1984; Lin and Watts, 1988), or that no definite bedding of volcanic rock can be obtained to constrain the paleomagnetic results. In fact, it is sometimes difficult to recover the original level of the vocanic bodies in Linzizong basin (Achache et al., 1984). For example, several paleomagnetic results from Tertiary volcanic rocks imply a paleolatitude of 12–22°N for the Lhasa area (Achache et al., 1984; Lin and Watts, 1988; Chen et al., 2010; Dupont-Nivet et al., 2010a; Liebke et al., 2010; Sun et al., 2010). On the contrary, Tan et al. (2010) reported a more northerly paleolatitude of 32°N for the Lhasa area.

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The amount of crustal shortening between Greater India and Eurasia due to the onset of India–Asia collision remains a subject of debate. Plate tectonic reconstructions based on paleomagnetic data from the Himalayan orogenic belt suggest that 2600–3400 km of post-collision convergence has taken place between Eurasia and greater India, with 1100–2000 km of this total length accommodated by north–south shortening in the Tibetan plateau and lateral tectonic escape (Achache et al., 1984; Patriat and Achache, 1984; Besse and Courtillot, 1991; Chen et al., 2010; Dupont-Nivet et al., 2010a; Liebke et al., 2010; Sun et al., 2010). However, recent paleomagnetic data suggest much lower amounts of continental shortening for the Late Cretaceous lava flows (200–600 km) and for the Eocene tuff (0 km) within Tibet than previously thought (Tan et al., 2010). The Himalayan shortening (south of Indus suture zone) is estimated by a mass-balanced cross-section of about 670 km from Pakistan to Sikkim (DeCelles et al., 2002). However, by balanced cross-section, Guillot et al. (2003) estimated a total Himalayan shortening of about 1100 km since 55 Ma. The difference between average values is explained by the existence of a Greater India that extended up to 650–700 km and the early subduction of the India plate north of the Indus suture zone. The crustal shortening in the Himalayas estimated by balanced cross-sections is much smaller than the value of crustal shortening implied by paleomagnetic investigations. Possible explanations regarding this problem include the inaccuracy of the APWP of the Eurasia and/or a non-dipolar field contribution (Dupont-Nivet et al., 2010b). Possible inclination shallowing is also a factor for the dispersion of data from sedimentary rocks (Tan et al., 2010).

In order to improve the paleolatitude of the southern margin of Eurasia, new paleomagnetic and geochronological data from the Late Cretaceous sediments in the Lhasa terrane are presented here which provide the necessary paleolatitude control on the southern margin of Eurasia block and constrain the magnitude of subsequent continental convergence.

2. Geological setting and sampling

The Lhasa terrane is the southernmost of several crustal blocks, which were accreted to the southern margin of Eurasia prior to the Cenozoic India–Asia collision (Fig. 1). Geological observations show the Lhasa terrane accreted onto the Qiangtang terrane of the southern Asia during the Late Jurassic–Early Cretaceous before the collision with India. The southern boundary of the Lhasa terrane marked by the Yarlung Zangbo Suture (YZBS) formed as a result of the Indo-Asian collision, which occurred between 70 and 38 Ma (Chang and Cheng, 1973; Yin and Harrison, 2000; Tapponnier et al., 2001).

The Shexing Formation exposed in the Lhasa terrane of southern Tibet (Fig. 1) overlies the early Cretaceous Takena Formation and is unconformably overlain by the early Tertiary volcanic strata of the Linzizong Formation (Fig. 2). In the study area (Maxiang Fig. 2), the Shexing Formation consists of red beds interbedded with andesite lava in the upper part of this studied section. Fossils identified in this formation include Ostracoda (*Altanicypris*, *Cypridopsis*) and spore–pollen (*Retitricolpites georgensis*, *Tricolpites*, *Cyathidites minor*, *C. australis*, *Densoisporites*, *Classopollis* sp., *Cerebropollenites* sp., *Biretisporites* sp.), indicative of Late Cretaceous time (BGMRXAR, 1993). In addition, U–Pb zircon ages of the intercalated andesite flows in the upper part of this studied section of 72.4 ± 1.8 Ma (see below) yield the minimum depositional age of the Shexing Formation. The youngest fossils in the Takena Formation are Albian (BGMRXAR, 1993). Therefore, the Shexing Formation lies within the time interval of ~72 and ~110 Ma. The Shexing Formation is overlain with an angular unconformity by the unfolded Linzizong Formation (Mo et al., 2007), thus indicating that the age of folding is older than Paleocene. The bedding of the Shexing Formation in the studied section dips at some 60–112° towards the SE. The bedding for the andesite lava flows were determined by the bedding of red beds interbedded in andesite lava in the studied section (Table 1).

Altogether we sampled 3 andesite lava sites and 17 Late Cretaceous red bed sites at Maxiang (29.9°N, 90.7°E) located ~80 km the northwest of Lhasa city (Fig. 2). Generally, 10 samples were taken at each site, using a gasoline-powered drill, and were oriented using solar and magnetic compass measurements.

In addition, one fresh andesite lava sample was collected at the upper part of the studied section to constrain the minimum depositional age of the Shexing Formation by U–Pb SHRIMP ages of zircons.

3. Age determination and paleomagnetic results

Dating of zircon was carried out using the SHRIMP II ion microprobe at the Beijing SHRIMP Centre, Chinese Academy of

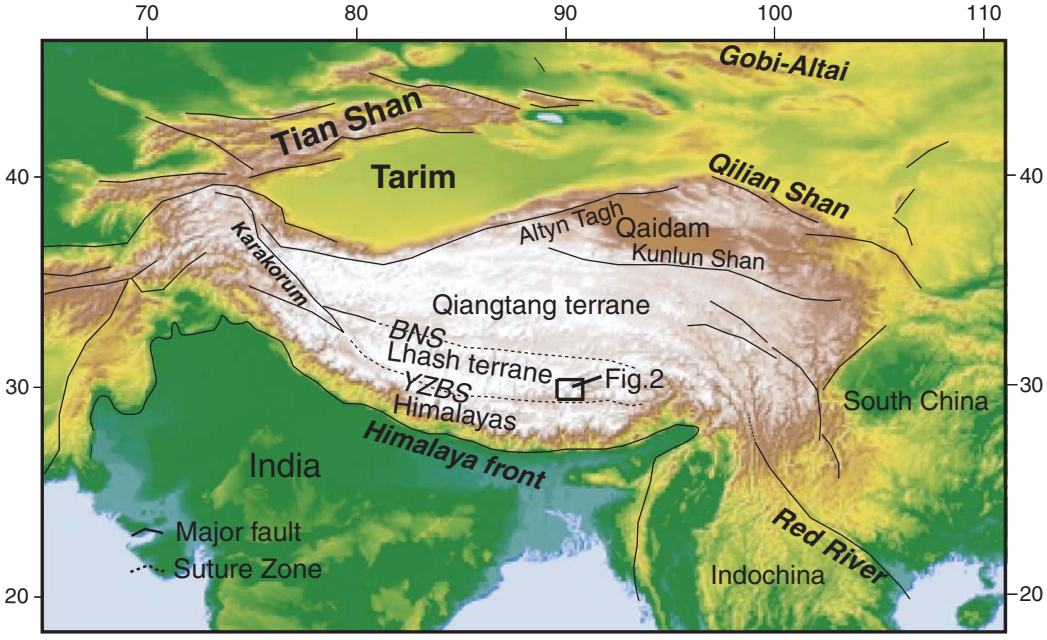


Fig. 1. Simplified tectonic map of Central Asia centered on Tibet, showing blocks and major faults.

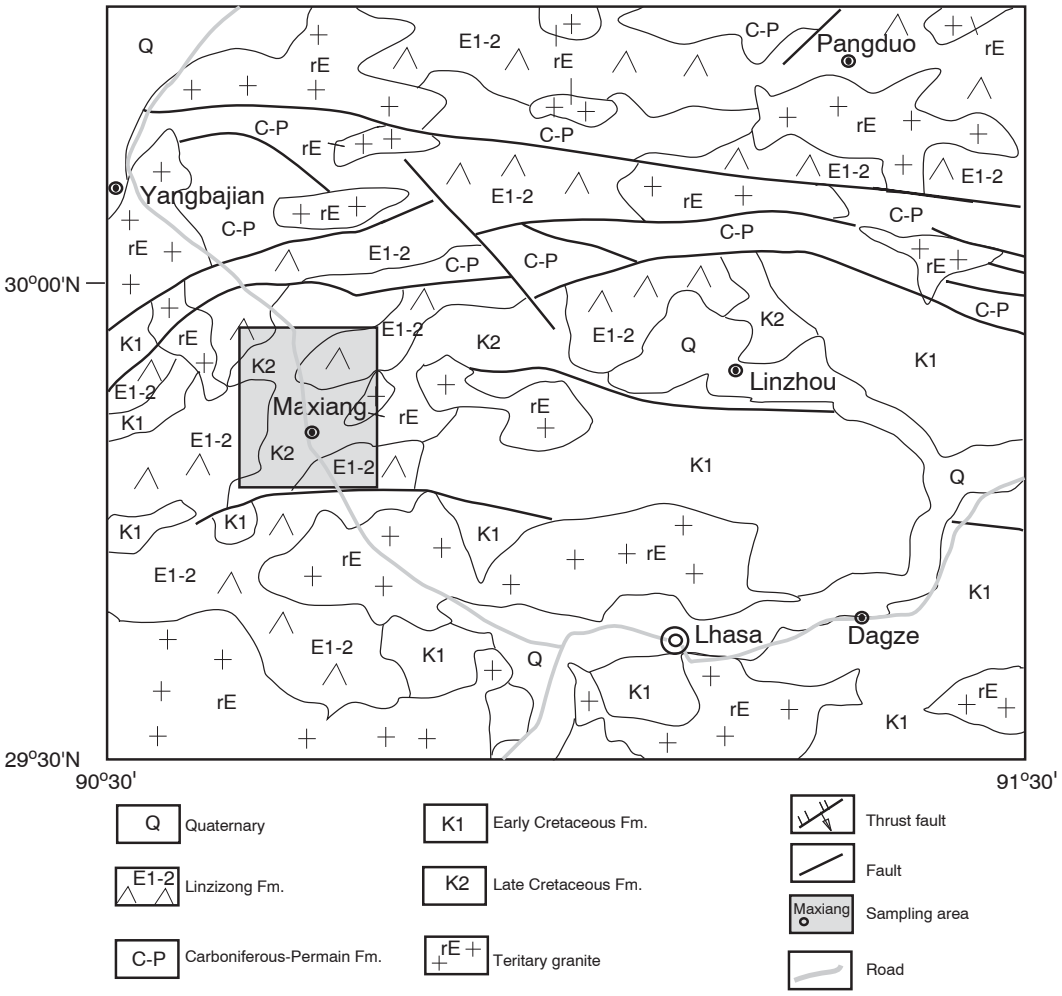


Fig. 2. Simplified geological map of the investigation area.

Geological Sciences CAGS. Analytical procedures for zircon were similar to those described by Williams (1998). Mass resolution during the analytical sessions was ~5000 (1% height). Spot sizes were ~30 μm and each site was rastered for 150 s before analysis. Data processing was carried out using the Squid and Isoplot programs (Ludwig, 2001), and measured ^{204}Pb was applied for the common lead correction. The results of the SHRIMP U–Pb analyses of zircons from sample XS-1 are presented in Table 1 and are plotted in Fig. 3, respectively. Grains have relatively high Th/U of ~0.51–1.13 (Table 1). All the separated zircon grains have typical igneous features. Seven analyzed grains define a concordant age of 72.4 ± 1.8 Ma (MSWD = 0.62) (Fig. 3) which we interpret to be the age of the andesite. This age most likely represent the minimum depositional age of the Shexing Formation, based on field appearance and occurrence.

All samples underwent stepwise thermal demagnetization using 14–19 steps in an ASC TD-48 oven with an internal residual field of less than 10nT. Remanent magnetization was measured with a 2G cryogenic magnetometer at the palaeomagnetic laboratory of the Institute of Geomechanics, CAGS in Beijing. Magnetization directions were determined by principal component analysis. The site-mean of paleomagnetic directions was calculated using Fisherian statistics.

The anisotropy of magnetic susceptibility (AMS) was measured to assess if these rocks have suffered substantive tectonic strain. Anisotropy of Magnetic Susceptibility (AMS) from the red beds indicates a clear sedimentary fabric with a nearly vertical minimum axis (Fig. 4a). A mean anisotropy degree (P) of 1.015 (range of 1.008 to 1.03) is revealed for the Late Cretaceous red beds (Fig. 4b). In the few studies that have performed AMS measurements, Thomas et al.

Table 1
Summary of SHRIMP U–Pb data on zircons for sample XS-1 from the Shexiang section.

Spot	% $^{206}Pb_c$	ppm U	ppm Th	Th/U	$^{207}Pb^*/^{206}Pb^*$	±%	$^{206}Pb^*/^{238}U$	±%	$^{206}Pb/^{238}U$ Age	$^{207}Pb/^{206}Pb$ age
XS-1-1	0.91	540	469	0.87	0.0485	6.8	0.01090	3.1	69.9 ± 2.1	123 ± 160
XS-1-2	3.16	95	49	0.52	0.053	42	0.01121	4.5	71.9 ± 3.2	345 ± 950
XS-1-3	0.69	626	700	1.12	0.0473	6.5	0.01156	3.0	74.1 ± 2.2	63 ± 160
XS-1-4	0.16	1636	2175	1.13	0.0480	2.5	0.01124	2.9	72.1 ± 2.1	97 ± 59
XS-1-5	2.53	100	60	0.60	0.064	26	0.01174	4.1	75.3 ± 3.0	742 ± 550
XS-1-6	3.72	129	66	0.51	0.0398	20	0.01160	3.6	74.3 ± 2.7	−363 ± 520
XS-1-7	1.52	238	181	0.76	0.0391	17	0.01110	3.3	71.2 ± 2.3	−410 ± 430

Pb_c and Pb* indicate the common and radiogenic portions, respectively.

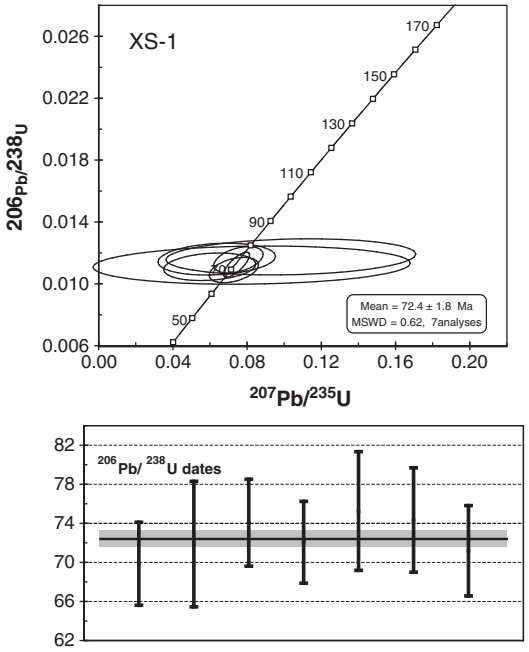


Fig. 3. Concordia plot of SHRIMP U–Pb data for zircons from andesite sample XS-1from the above part of Shexing Formation; Gray bars in lower panel shows weighted mean $^{206}U/^{238}Pb$ ages; black columns represent single analyses.

(1993) found negligible to low (generally <5%) AMS for inclination shallowing in sedimentary rocks. However, other studies suggest that measurements of AMS are of limited value in assessing whether or not inclination shallowing has occurred within the sedimentary package (Tan et al., 2003, 2010). Based on the lower degrees (mean <1.015) of anisotropy and the large scatter in the maximum susceptibility axis directions (Fig. 4), we suggest that the Shexing Formation has not experienced significant strain due to compaction or tectonic stress.

Fig. 5 shows representative Zijderveld plots and normalized intensity curves. Most samples for the red beds posses a high-temperature component (HTC; 260–680 °C) which trends toward the origin. The high unblocking temperature (680 °C) indicates the presence of hematite for the red bed and andesite sites. The late Cretaceous HTC data for the red beds are listed in Table 2. The site mean HTC direction for red beds is Dg = 342.6°, Ig = −10.1°, kg = 5.5 (α95 = 16.7°, n = 17, in geographic coordinates) (Ds = 344.1°, Is = 34.5°, ks = 47.8, α95 = 5.2°, n = 17, after tectonic correction) (Fig. 6a, Table 2), the corresponding paleopole lies at 71.9°N, 327.2°E with A_{95} = 4.5°. As listed in Table 2, The HTC directions after bedding correction show only normal polarity observed in all sedimentary sites, and The HTC directions in geographic coordinates are significantly different from the present-day field direction. This fact is consistent with deposition and magnetization within the Cretaceous Long Normal Superchron with an age between 118 and 83 Ma. The unfolding process reveals that the sites mean direction after tilt correction gives an optimal untilting at about 96.1% unfolding (with uncertainties ranging from 90.7 to 101.6%) (D = 344.0°, I = 32.9°) (Watson and Enkin, 1993), which is statistically indistinguishable from that of 100% unfolding (D = 344.1°, I = 34.5°, a_{95} = 5.2°). In addition, the McFadden (1990) correlation fold tests yield ξ1,g = 8.3 and ξ1,s = 0.58, in geographic and stratigraphic coordinates, respectively. The critical values are 4.8 and 6.72 at 95% and 99% confidence levels, respectively. Therefore, the McFadden fold test is positive. Thus we consider that the HTC magnetization of Shexing Formation red beds at 100% unfolding may represent the primary remanence acquired in the late Cretaceous.

Thermal demagnetization isolated a single dominant magnetic component in most andesite samples after removing a weak viscous magnetization (0 °C–500 °C) (Fig. 5g–h). In general, a high unblocking temperature component (HTC) is often rather easily isolated between 500 °C and 680°. The demagnetization behavior of the andesite sites is not similar to that of the red beds (Fig. 5a–f). The red bed samples exhibited a wide range of unblocking temperatures (260 °C–680 °C). However, this andesite sites exhibited linear demagnetization trend to decay toward the origin in a narrow range of unblocking temperatures (500–680 °C) (Fig. 5g–h).

If we combine the andesite sites with the sedimentary sites, the fold test is also positive. The McFadden (1990) correlation fold tests yield ξ1,g = 8.16 and ξ1,s = 2.22, in geographic and stratigraphic coordinates, respectively. The critical values are 5.2 and 7.3 at 95% and 99% confidence levels, respectively. So, we suggest that the remanence is likely primary. As listed in Table 1, the tilt-corrected mean direction from the basalt sites plus baked red bed site is D_s = 350.8°, I_s = 32.1°, κ_s = 17.3, α_{95} = 8.1°, and N = 20 sites, which corresponds to a pole at 75.0°N, 306.7°E with A_{95} = 6.8°(Fig. 6b, Table 2).

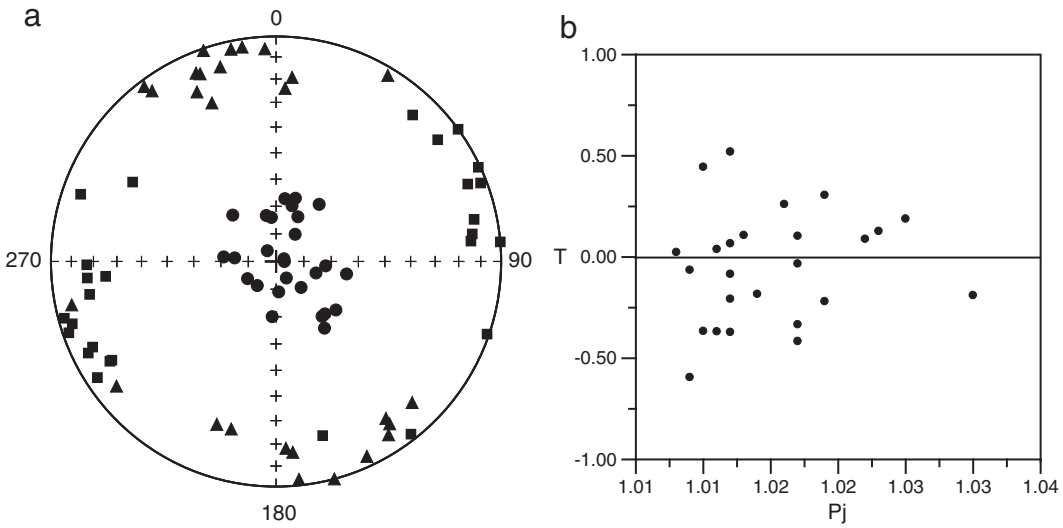


Fig. 4. (a) Anisotropy of magnetic susceptibility data in stratigraphic coordinates from the Shexing formation rocks. Lineation = k1/k2, Foliation = k2/k3. (b) Plots of shape parameter (T) versus anisotropy degree (P′).

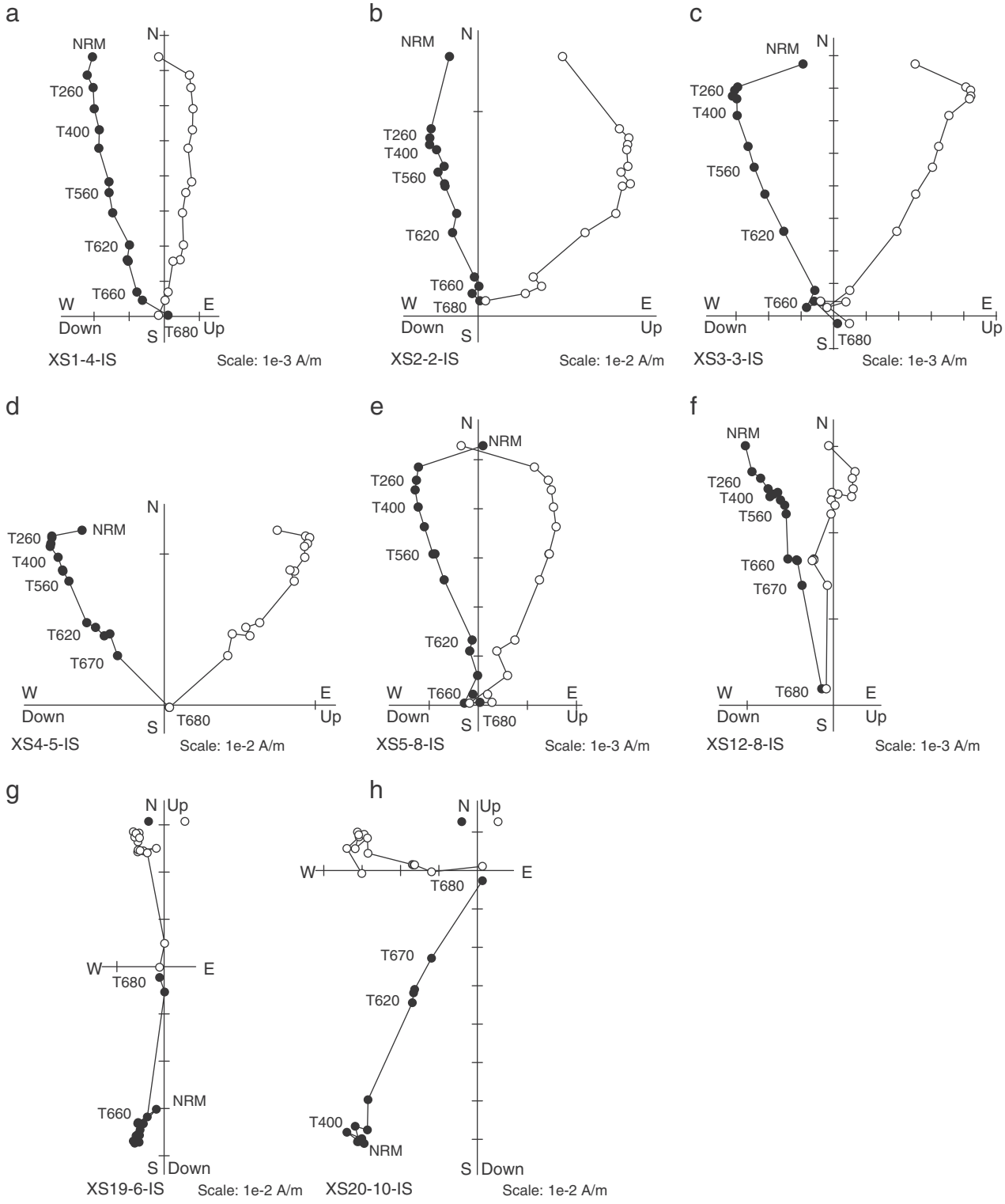


Fig. 5. Orthogonal demagnetization diagrams of representative rocks (sample number and corresponding sites listed) in situ coordinates. Demagnetization steps in °C in all plots. NRM is the natural remanent magnetization. (a–f) Typical examples from red beds; (g–h) Typical examples from andesite.

4. Results and discussion

4.1. Palaeolatitude of the southern margin of Asia between early Cretaceous and Paleocene

Previous palaeomagnetic investigations on rocks of the Cretaceous volcanic sequences (114.2 ± 1.1 Ma) in the eastern Lhasa block in localities near Deqing indicates a palaeolatitude of 14.0 ± 6.9°N for the southern margin of Asia (reference point: 29.9°N, 90.7°E) (Sun et al., 2008). The available late Cretaceous studies from the Lhasa block are listed in Table 3, and the corresponding paleopoles are shown in Fig. 7a.

The mid-Cretaceous sedimentary Shexing Formation (mostly red beds) indicate paleolatitudes (for the reference point) of ~7°N (Lin and Watts, 1988), ~11°N (Achache et al., 1984), ~21°N (Westphal and Pozzi, 1983), and ~12°N (Tan et al., 2010). Possible inclination shallowing is a critical cause for the validity of data from sedimentary rocks. Tan et al.

Table 2
Site-mean paleomagnetic directions of the high-temperature from Late Cretaceous rocks in Maxiang area (29.9°N, 90.7°E), Northwestern Lhasa.

Site	Slat	Slong	n/N	D_g	I_g	D_s	I_s	κ	α_{95}	Strike	Dip
Red beds											
XS1	29°56.021′	90°43.58′	10/10	355.0	−32.5	354.7	30.5	8.3	17.8	69	61
XS2	29°56.021′	90°43.58′	8/8	342.3	−23.3	342.6	45.1	27.7	10.7	73	69
XS3	29°55.929′	90°43.594′	6/8	335.4	−27.4	335.4	27.5	174.4	5.1	70	67
XS4	29°55.929′	90°43.594′	5/7	335.3	−35.3	335.2	19.7	87.9	8.2	65	55
XS5	29°55.935′	90°43.594′	7/7	345.6	−29.1	345.2	25.0	50.3	8.6	64	55
XS6	29°55.935′	90°43.594′	4/6	337.5	−8.8	340.9	46.6	13.3	26.2	60	56
XS7	29°55.935′	90°43.594′	7/10	343.1	−18.6	345.4	35.9	38.6	9.8	60	56
XS8	29°55.935′	90°43.594′	7/9	355.2	−10.8	2.6	39.2	27.8	11.7	60	56
XS9	29°55.935′	90°43.594′	7/7	346.1	−17.2	349.6	41.4	36.1	10.2	64	60
XS10	29°55.935′	90°43.594′	7/7	338.8	−48.1	339.7	20.8	53.2	8.4	72	69
XS11	29°55.935′	90°43.594′	9/9	336.1	−33.4	336.2	34.5	46.3	7.6	63	68
XS12	29°52.956′	90°43.651′	9/9	356.6	12.4	353.8	27.6	44.7	7.8	112	17
XS13	29°52.956′	90°43.651′	6/7	349.8	12.1	346.5	26.3	16.8	16.8	112	17
XS14	29°52.956′	90°43.651′	4/5	348.3	−2.9	339.1	49.0	17.8	22.4	292	173
XS15	29°50.092′	90°43.601′	5/7	340.1	−1.9	327.5	46.4	10.8	24.4	93	55
XS16	29°53.135′	90°43.636′	6/8	302.7	79.6	354.7	34.8	31.0	12.2	276	50
XS17	29°53.135′	90°43.636′	4/6	280.2	66.4	339.3	31.0	86.8	9.9	276	50
Mean	29°53.135′	90°43.636′	17(111)	342.6	−10.1			5.5	16.7		
						344.1	34.5	47.8	5.2		
Corresponding pole: 71.9 N/327.2E, $d_p/d_m = 3.4/6.0$, $A_{95} = 4.5$											
Andesite											
XS18	29°53.508′	90°43.526′	5/6	189.4	−31.5	217.1	−14.9	49.1	11	279	60
XS19	29°53.619′	90°43.499′	5/6	196.8	−60.7	192.8	−0.9	137.2	6.6	279	60
XS20	29°53.678′	90°43.483′	5/5	248.3	−71.7	205.6	−19.6	26.1	15.2	279	60
Mean	Red beds + Andesite		20(126)	346.3	−0.5	–	–	3.9	19.2		
				–	–	350.8	32.1	17.3	8.1		
Corresponding pole: 75.0 N/306.7E, $d_p/d_m = 5.1/9.1$, $A_{95} = 6.8$											

Slat and slong: latitude and longitude of the sampling site; n/N: number of samples used to calculate mean and measured; D_g, I_g, D_s, I_s: declination and inclination in geographic and stratigraphic coordinates respectively; κ: the best estimate of the precision parameter; α95: the radius that the mean direction lies within 95% confidence. Strike: dip direction of beds−90°, Dip: the mean dip and dip azimuth of site beds; unit: degree.
(1) Fold test for red beds is positive according to.
McFadden (1990) method: critic Xi at 95% = 4.8, at 95% = 6.7, Xi1 IS = 8.3, Xi1 TS = 0.58.
(1) Fold rest for red beds + Andesite is positive according to.
McFadden (1990) method: critic Xi at 95% = 5.2, at 95% = 7.3, Xi1 IS = 8.16, Xi1 TS = 2.22.

(2010) calculate a corrected paleolatitude of ~22°N by the E/I analysis (Tauxe and Kent, 2004). This implies that the Cretaceous red beds suffered from inclination shallowing (10° less than lava flows), and inclination correction yielded consistent paleolatitude data for the red beds and volcanic rocks. The data in their study (Tan et al., 2010) are carefully analyzed and useful. However, although the test is valid to identify the existence of inclination shallowing, one should be careful with interpreting corrected results quantitatively (Najman et al., 2010). Further available Cretaceous data from the volcanic rocks yield paleolatitudes of 13°N–18°N (Lin and Watts, 1988; Sun et al., 2008).

This streaking of late Cretaceous poles from the Lhasa terrane discourages us from using the average of currently available poles as a reference for the entire Lhasa terrane. It is apparent in Fig. 7a that the late Cretaceous poles (lava flows) from the Linzhou area (Tan et al., 2010) on the Lhasa terrane lie far east from other areas for the Lhasa terrane. However, the late Cretaceous poles from three studies (this study; Pozzi et al., 1982; Westphal and Pozzi, 1983) are completely overlapped at the 95% confidence level. Apart from the late Cretaceous pole from Tan et al. (2010), the other late Cretaceous poles from similar regions in the Lhasa terrane (Maxiang, Linzhou) align on a well-defined small circle centered on the reference point (29.9°N, 90.7°E) (Fig. 7a). This indicates a constant paleolatitude for the Lhasa terrane during the late Cretaceous. The small circle that passed over the six poles (this study; Pozzi et al., 1982; Westphal and Pozzi, 1983; Achache et al., 1984; Lin and Watts, 1988; Tan et al., 2010) fits at a reference point near Maxiang area (29.9°N, 90.7°E) with a colatitude of 74.5° ± 5.7°. The paleolatitude of 15.5° ± 5.7°N for the Lhasa terrane was calculated at the same reference point (see Table 4 and Fig. 7a).

If we compare the observed paleolatitude with expected values from the adjacent blocks, the Lhasa terrane shows significant relative north–south displacement (10.4 ± 8.2° and 19.4 ± 6.7°) since Late Cretaceous with respect to Eurasia (Besse and Courtillot, 2002) and Tethyan Himalaya (Patzelt et al., 1996) respectively within paleomagnetic uncertainties.

Previous palaeomagnetic investigations on rocks of the Paleocene Linzizong Formation yield palaeolatitudes of 32.7 ± 4.8° N (Tan et al., 2010), 7.4–10.2° N (Chen et al., 2010), 22.8 ± 3.3° N (Dupont-Nivet et al., 2010a), 12.7° ± 6.3° N (Achache et al., 1984), 13.2–13.6°N (Liebke et al., 2010; Sun et al., 2010) and 8.3° ± 16.4° N (Westphal and Pozzi, 1983) for the southern margin of Eurasia. These results are not consistent because of too few sites, partial overprints or the spot recording of the geomagnetic field in some studies. Nevertheless, as seen in Fig. 7b, all the presently available results align on a well-defined small circle centered on the reference point (29.9°N, 90.7°E) except that one of Tan et al. (2010). This indicates a constant paleolatitude for the Lhasa block during the Paleocene period. The small circle that passed over the six poles (Achache et al., 1984; Chen et al., 2010; Dupont-Nivet et al., 2010a; Liebke et al., 2010; Sun et al., 2010) is fit at a reference point near Maxiang area (29.9°N, 90.7°E) with a colatitude of 74.5° ± 5.7°. The Paleocene paleolatitude of 15.5° ± 5.7° for the Lhasa terrane was calculated at the same reference point (see Table 4 and Fig.7b). Our result places the southernmost margin of Asia at about 15°N between late Cretaceous and 55 Ma (for a reference point: 29.9°N, 90.7°E) (Fig. 8, Table 4). This indicates that the southern margin of the Lhasa terrane was rather straight before India–Asia collision. Conformity of Eocene and Cretaceous data indicates that no larger latitudinal motion of the Lhasa terrane occurred between Cretaceous and Paleocene.

On the other hand, it is apparent in Fig. 7 that both late Cretaceous and Paleocene poles from the Lhasa block lie far away from those for the stable India and Eurasia block. We combine available results from upper Cretaceous sediments of the Lhasa terrane (this study; Pozzi et al.,

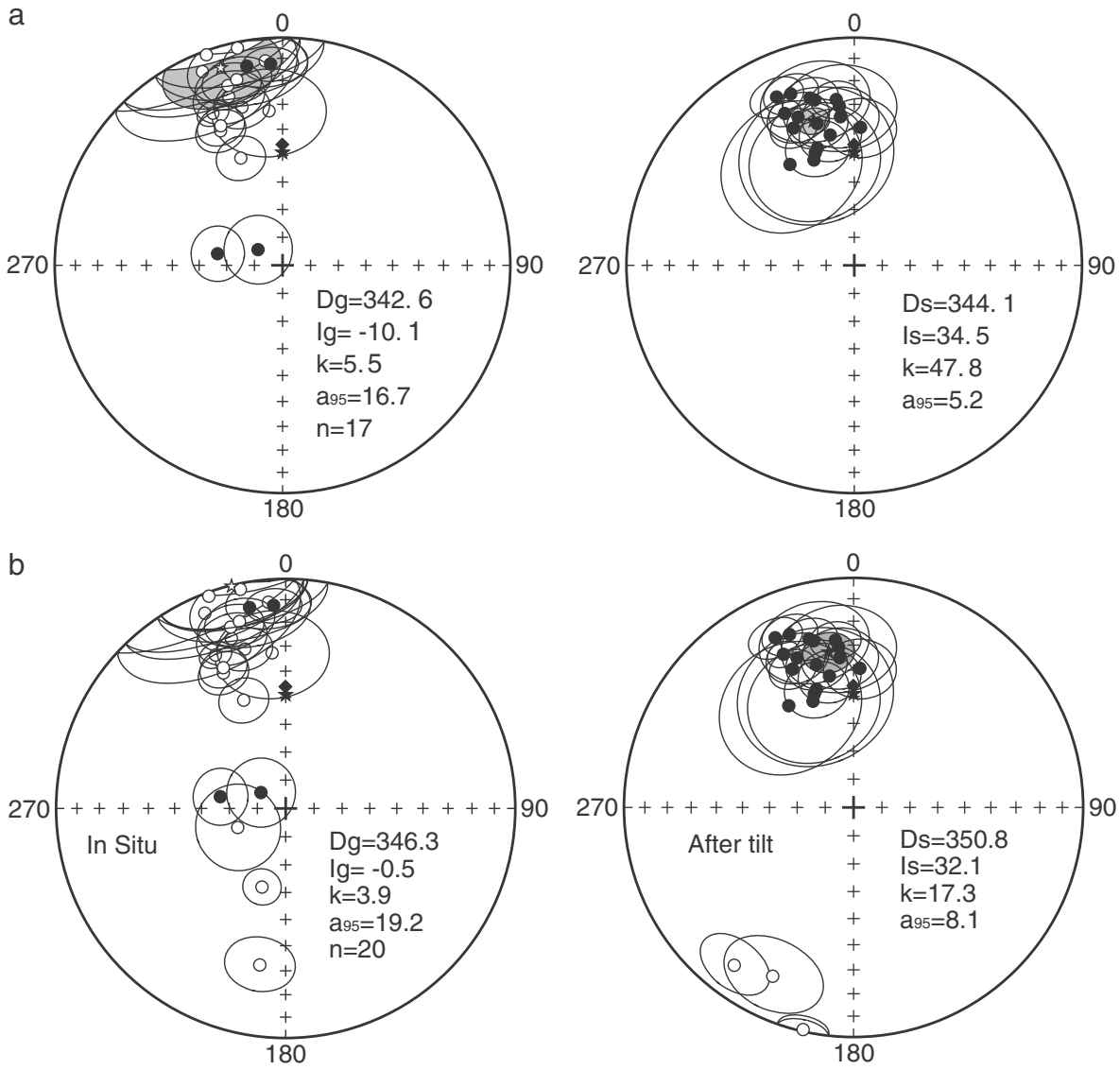


Fig. 6. Equal-area projections of the high temperature component for sediments (a) and all sites including volcanic rocks (b).

1982; Westphal and Pozzi, 1983; Lin and Watts, 1988; Achache et al., 1994; Tan et al., 2010) to calculate a Cretaceous latitude of the southern margin of Asia of 15.5 ± 5.7°N (see Table 4 and Fig. 8). The palaeolatitude for the southern margin of Eurasia calculated from the apparent polar wander path is 25.9 ± 5.9° (Besse and Courtillot, 2002), indicating that the latitudinal distance of the suture with respect to Eurasia during the late Cretaceous was 1100 ± 860 km (Table 4 and Fig. 7a). This result is little less than the Palaeocene results from the Linzizong Formation (1700 ± 800 km; see Tables 4 and Fig. 7b). This result, however, does imply that most of the intra-Asian convergence occurred after the collision and that pre-collisional intra-Asian shortening is limited to within palaeomagnetic uncertainties.

The collision palaeolatitude at 15.2 ± 6.3°N further implies that the total latitudinal convergence was accommodated by 1700 ± 800 km (16.2 ± 7.6°) between southern Tibet and stable Eurasia and 1500 ± 830 km (14.4 ± 7.9°) between southern Tibet and stable India (Fig. 8, Table 4). Such large amount of N–S convergence referred by the paleomagnetic data have been previously accounted for by a combination of internal deformation (folding and thrusting), and large lateral displacements (extrusion) along the Altyn Tagh and red River faults (Dewey et al., 1989; Yang and Besse, 1993; Jin et al., 1996; Yin and Harrison, 2000; Yang et al., 2001; Chen et al., 2002; Charusiri et al., 2006). This shortening may result from subduction of a limited amount of continental crust along the Yarlung Zangbo suture zone and from underthrusting of continental crust along major thrust faults such as the main Boundary Thrust (MBT), main Central Thrust (MCT), and Kangmar thrust, south of the Yarlung Zangbo suture (Zhao et al., 1993; Yin and Harrison, 2000; DeCelles et al., 2002; Royden et al., 2008).

The Cenozoic Eurasia reference poles predict intracontinental convergence values (more than 1000 km within uncertainty) that are irreconcilable with the geological record of upper crustal shortening in Asia. Previous studies (e.g. Cogn´e et al., 1999; Tan et al., 2003; Dupont-Nivet et al., 2010b; Lippert et al., 2010; Najman et al., 2010) proposed several hypotheses to explain this inclinational discrepancy such as compaction-induced inclination shallowing, non-dipolar geomagnetic field geometry, tectonic shortening, or a poorly constrained APWP for Eurasia and or non-rigidity of the Eurasian plate.

The APWP for Eurasia (Besse and Courtillot, 2002) is usually employed as a reference curve for late Mesozoic–Cenozoic paleomagnetic studies in Asia. As we know, the Late Cretaceous and Paleogene reference poles for Eurasia are based mainly on data from northwestern Europe (e.g., Germany, United Kingdom, and France),

Table 3
Paleomagnetic poles from Lhasa terrane and from adjacent blocks (reference point: 29.9°N, 90.7°E).

Block or areas	Age/(Ma)	Rock type	Lat.	Lon.	N	Plat.	Plon.	A_{95} (dp/dm)	Reference
Lhasa Block	~114	Rhyolitic tuff	30.5	90.1	15	66.4	220.3	6.9	Sun et al. (2008)
	K2	Sediments	30.0	91.0	8	71.2	288.4	7.9	Achache et al. (1984)
	K2	Sediments	29.4	91.0	7	68.0	340.0	6.0	Pozzi et al. (1982)
	K	Sediments	33.7	80.4	14	66.2	245.0	5.1	Chen et al. (1993)
	K2	Sediments	29.9	90.7	17	71.9	327.2	4.5	This study
	K2	Sediments + andesite	29.9	90.7	20	75.0	306.7	6.8	This study
	K2	Sediments	29.9	91.0	6	64.0	348.0	5.6/9.5	Westphal and Pozzi (1983)
	K2	Sediments	29.9	91.2	8	68.0	279.0	3.5/6.9	Lin and Watts (1988)
	K2	Sediments	29.9	91.2	43	70.2	300.5	1.9	Tan et al. (2010)
	K2	Lava flows	29.9	91.2	21	69.1	191.7	4.2	Tan et al. (2010)
	45–65	Volcanic rocks	30.0	91.0	8	71.4	299.8	11.0	Achache et al. (1984)
	45–65	Volcanic rocks	30.0	91.1	37	81.2	221.4	4.2	Dupont-Nivet et al. (2010a, 2010b)
	45–65	Volcanic rocks	30.0	91.2	10	87.5	81.4	5.9	Tan et al. (2010)
	64–60	Volcanic rocks	29.9	90.2	15	66.0	284.9	8.5	Chen et al. (2010)
	60–44	Volcanic rocks	29.9	90.2	23	70.6	281.0	5.3	Chen et al. (2010)
	~55	Rhyolitic tuffs	30.1	90.9	14	73.6	274.3	7.3	Sun et al. (2010)
	~53	Dykes	30.1	90.9	10	69.0	224.3	7.8	Liebke et al. (2010)
Tethyan Himalaya	55–63	Limestones	28.3	88.5	14	68.2	277.1	3.1	Patzelt et al. (1996)
(Gamba)	65–71	Limestones	28.3	88.5	14	55.8	261.6	3.5	Patzelt et al. (1996)
India APWP	55					60.7	277.7	4.8	Besse and Courtillot (2002)
	80					29.2	284.7	6.7	Besse and Courtillot (2002)
Europe APWP	55					81.4	168.3	4.2	Besse and Courtillot (2002)
	80					81.4	206.1	5.9	Besse and Courtillot (2002)

Lat. (Lon.): latitude (longitude) of sites; Plat (Plon): latitude (longitude) of pole; N: sites (studies); A_{95}/(dp/dm): the radius that the mean pole lies within 95% confidence; K2: Late Cretaceous. unit: degree.

with few from Siberia or eastern Asia. On the other hand, recent geological and geophysical evidence suggests that the European platform has not behaved as a single stable plate since the Late Cretaceous (Yegorova and Starostenko, 1999; Trifonov, 2004).

Inclination shallowing was thought to best explain this discrepancy between the central Asian and Eurasian, which has since been corroborated using numerical techniques and by comparing the sedimentary paleomagnetic data with those from volcanic rocks (Bazhenov et al., 2002; Gilder et al., 2003; Tan et al., 2003; Tauxe and Kent, 2004; Huang et al., 2005). However, some paleomagnetic results imply no difference in inclination between sediments and volcanic rocks (Thomas et al., 1993; Bragin et al., 1999; Sun et al., 2006). In addition, inclination shallowing poorly explains why Cenozoic inclinations are systematically shallower than the Cretaceous ones in the central Asia despite being derived from similar rock types. Thus it is unclear that inclination shallowing is the principal reason for the discrepancy between the central Asian and Eurasian Late Jurassic to present paleomagnetic poles. However, further Cretaceous–Tertiary paleomagnetic studies on the red beds, especially on the volcanic flows are needed in order to better resolve the problem of inclination shallowing in Lhasa terrane.

4.2. Timing of Indo-Asia collision

To estimate the age of the onset of collision, we assess the timing at which the palaeolatitude of the southern margin of Asia as defined by our results overlaps with the palaeomagnetically determined palaeolatitude estimates from the Tethyan Himalayan sediments taken to represent the northern extent of Greater India.

Fig. 8 displays the expected palaeolatitude reference curves of India and Eurasia as well as palaeolatitudes calculated from present and previous palaeomagnetic investigations. Expected palaeolatitude reference curves were established from apparent polar wander paths (APWP) for stable India and Eurasia (Besse and Courtillot, 2002) based on the present day geography with a reference point at the YZBS (29.9°N, 90.7°E).

The palaeolatitude of the northern Greater Indian margin is calculated from palaeomagnetic data collected directly south of the Indo-Asia suture from sites at Zongpu (63–55 Ma) and at Zongshan (71–65 Ma) (Patzelt et al., 1996). The youngest of these data sets indicates that the Tethyan Himalayan sediments were located 7.4 ± 3.1°N at 59 ± 4 Ma, clearly much farther south than the southern margin of Asia (Fig. 8 and Table 4) (Patzelt et al., 1996). Thus, the collision between the Tethyan Himalayas and the Lhasa Block could not have occurred before 59 ± 4 Ma (Dupont-Nivet et al., 2010a; Liebke et al., 2010). There is also the possibility that Greater India was extended further north of the sampling sites of Patzelt et al. (1996) (Ding et al., 2005). Dupont-Nivet et al. (2010a, 2010b) derive a slightly younger age of 48 Ma (57–40 Ma within 95% confidence limits) and Chen et al. (2010) calculate a collisional time of 65–50 Ma (or narrowed down to 60–55 Ma assuming a 950 km extent of Greater India). Liebke et al. (2010) suggested an age for the India-Asia collision between 53 and 49 Ma with a 95% confidence limit of ±6 Ma. These results are all in agreement with the generally considered age of the India-Asia collision (~55 Ma) by the geological records. However, the paleomagnetic results from the Pana Formation (Tan et al., 2010) give a younger collision age of 43 Ma. Caution should be taken when considering whether the sampled volcanic tuffs could represent a short-term large eruption event potentially leading to a snapshot record of the Earth magnetic field at high inclination (Dupont-Nivet et al., 2010b).

The age of initial collision between India and Eurasia can be determined by intersection of the pre-collisional southern margin of Eurasia and northern margin of India. To estimate the age of the collision, we thus extrapolate the northward path of India through the palaeolatitudes of the Tethyan Himalayas until they intersect with the palaeolatitudes of the southern Lhasa margin at 15.2 ± 6.3°N (Fig. 8). This provides a 47 Ma minimum age for the collision with 95% confidence interval between 47 Ma and 54 Ma (Fig. 8 and Table 4). The palaeomagnetic results may preclude the possibility that collision began after 35 Ma at the longitude of Lhasa (Aitchison et al., 2007).

Defining the onset of the India–Asia collision depends on knowing the paleogeographic plate boundaries for Eurasia and Greater India. As our study area lies at the Asian southern margin prior to collision, the paleolatitude derived from the Lhasa block provides a key constraint on the age of the India–Asia collision as well as the amount of intra-plateau deformation since the collision. Paleolatitudes based on both the Late Cretaceous sedimentary data and the Paleocene volcanic rocks placed the Lhasa block at about 15°N. Our new data imply that at 55 Ma, the greater India needed to move 7° farther north before

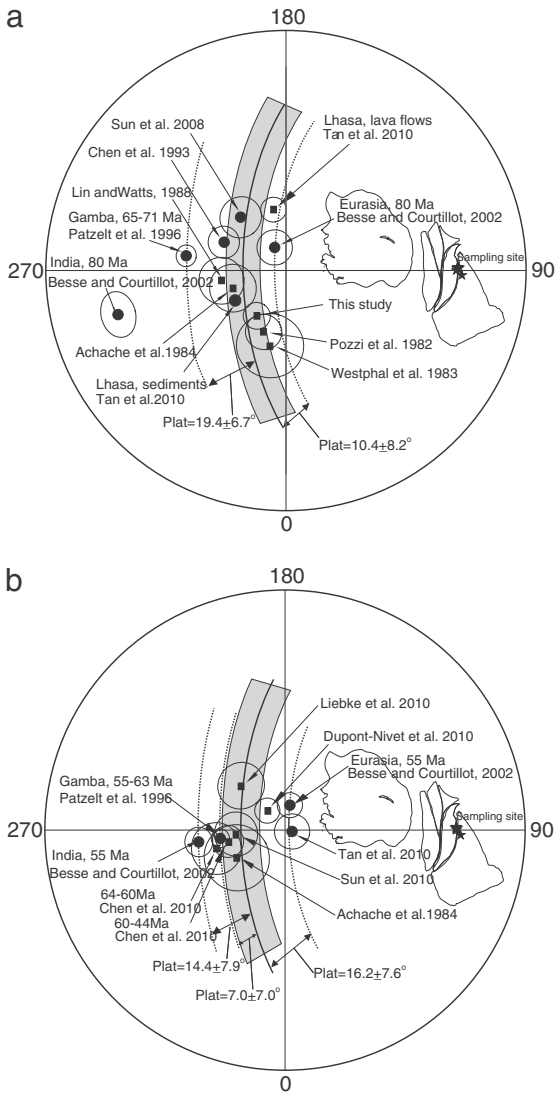


Fig. 7. Equal-area projection of late Cretaceous poles (a) and Paleocene (b) of the Lhasa terrane and adjacent blocks. See Table 4 for values and references. Dotted line: small-circle running through reference pole of India (Eurasia) centered on our site locations; solid lines: small-circles passing through our site locations; stars: sampling location. Square: the poles used for the fit of the small circle; plat = latitudinal displacement; light gray area with bold black line in (a) and (b): small-circle with its 95% confidence area passing through palaeopoles, and centered on the average site locations.

reaching the southernmost margin of Asia, which, using a mean northward drift velocity of ~92 mm/yr for India (Lee and Lawver, 1995), requires 8 million years. Therefore, our favored initial collision age at the longitude of Lhasa is 47–54 Ma. An initial collision within this time range by the palaeomagnetic constraints is in accordance

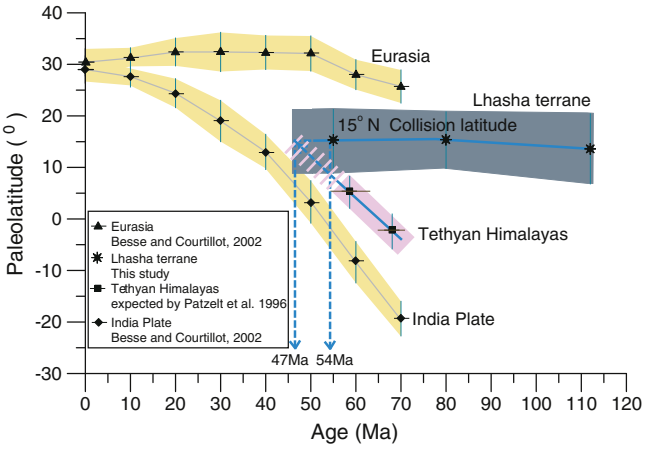


Fig. 8. Plot of palaeolatitudes of the Eurasian and Indian continents and the pre-collisional northern margin of India and southern margin of Eurasia. The expected palaeolatitudes calculated from APWPs (Patzelt et al., 1996; Besse and Courtillot, 2002) for a reference point at the reference point (29.9 N°/90.7°E). The India and Eurasia latitudinal paths with 95% confidence interval (shaded yellow areas), vertical bars display the errors of the palaeolatitude. The age of the collision is estimated using the intersection between the palaeolatitude of the southern margin of the Lhasa terrane and that of the Tethyan Himalayas (Table 4), the correlated India–Asia collision times are marked by arrows.

with previous studies based on geologic data (Guillot et al., 2003, 2007, 2008; Green et al., 2008; Copley et al., 2010).

5. Conclusion

New paleomagnetic results from 20 sites of the late Cretaceous Shexing Formation were obtained near Maxiang in the Lhasa Terrane. A high temperature characteristic magnetic component is isolated and passes a fold test. We argue that this is a primary magnetization which can be dated between 110 and 72 Ma. Based on presently available results from the Lhasa terrane, the southernmost margin of Asia was located at about 15°N between Late Cretaceous and 55 Ma (for a reference point: 29.9°N, 90.7°E). This indicates that the southern margin of the Lhasa terrane was rather straight before India–Asia collision. Conformity of Paleocene and Cretaceous data indicates that no larger latitudinal motion of the Lhasa terrane occurred between Cretaceous and Paleocene. When the results are compared to the studies available from the adjacent blocks, new paleomagnetic data from the Lhasa block imply that the total latitudinal convergence was accommodated by 1700 ± 800 km (16.2 ± 7.6°) between southern Tibet and stable Eurasia and 1500 ± 830 km (14.4 ± 7.9°) between southern Tibet and stable India. A collision age (54–47 Ma) was determined using the results for the southernmost margin of Asia according to our new data and that from 'Greater India' (Patzelt et al., 1996). A much younger age for the initial India–Asia collision as assumed by Aitchison et al. (2007) and Tan et al. (2010) is clearly disproved.

Table 4
Paleomagnetic data constraining the paleolatitude of the Lhasa terrane during Early Cretaceous to Paleocene (reference point: 29.9°N, 90.7°E).

Block	Age (Ma)	Observed plat.	Relative to Eurasia		Relative to Tethyan Himalaya		Relative to India		References
			Expected plat.	△plat	Expected plat.	△plat	Expected plat.	△plat	
Lhasa Block	K2	15.5 ± 5.7	25.9 ± 5.9	10.4 ± 8.2	−3.9 ± 3.5	19.4 ± 6.7	−29.4 ± 5.9	44.9 ± 8.8	1, 2, 3, 4, 5, 6,
	~55	15.2 ± 6.3	31.4 ± 4.2	16.2 ± 7.6	8.2 ± 3.1	7.0 ± 7.0	0.8 ± 4.8	14.4 ± 7.9	2, 7, 8, 9, 10
	K1	14.0 ± 6.9	–	–	–	–	–	–	11

△plat is derived from observed paleolatitude minus expected paleolatitude at locality (reference point: 29.9°N, 90.7°E) calculated from reference pole. The observed paleolatitude was calculated by the small circle that passed over those poles fitting at the same reference point near Maxiang; Ref.—reference of data : 1—This study; 2—Achache et al., 1984; 3—Pozzi et al., 1982; 4—Westphal et al. 1983; 5—Lin and Watts, 1988; 6—Tan et al., 2010;7—Dupont-Nivet et al., 2010a, 2010b; 8—Chen et al., 2010; 9—Sun et al., 2010; 10—Liebke et al., 2010; 11—Sun et al., 2008. unit: degree.

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